AI-Based Crop Disease Detection and Recommendation System Using Deep Learning for Smart Agriculture

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Type of Project
Product-Based Project
Domain: Artificial Intelligence, Computer Vision, AgriTech, and Smart Agriculture

Introduction

Agriculture plays an important role in the economy and food security, but crop diseases and nutrient deficiencies can significantly reduce crop production and farmers' income. Early identification of diseases is essential for preventing their spread and reducing crop losses. However, traditional disease detection methods often depend on manual observation and expert knowledge, which can be time-consuming, difficult, and sometimes inaccurate.

The proposed AI-Based Crop Disease Detection and Recommendation System uses Computer Vision and Deep Learning to automatically analyze images of crops and leaves captured using a mobile phone. The system processes the image and identifies visual symptoms such as spots, discoloration, patterns, and texture changes. Based on the analysis, the system predicts the possible disease or crop stress and provides suitable recommendations to the farmer.

The system can also consider environmental conditions such as temperature and humidity to improve prediction accuracy. By providing quick and accessible disease detection, the project aims to support farmers in taking timely action and improving crop productivity.

Problem Statement

Farmers often face difficulties in identifying crop diseases at an early stage. Diseases may initially appear as small spots, colour changes, or other symptoms that can be difficult to recognize without expert knowledge. Delayed or incorrect identification can result in disease spreading, excessive use of pesticides, crop damage, and financial losses.

Existing manual inspection methods require considerable time and may not always be available to farmers, especially in rural areas. Therefore, there is a need for an intelligent, affordable, and user-friendly system that can automatically identify crop diseases from images and provide appropriate recommendations.

Objective

The main objectives of the project are:

- To develop an AI-based system for detecting crop diseases using plant or leaf images.

- To apply Deep Learning and Computer Vision techniques for accurate image-based disease classification.
- To preprocess crop images by removing noise, adjusting brightness, resizing, and enhancing relevant features.
- To provide the predicted disease name along with a confidence score.
- To consider environmental factors such as temperature and humidity for improved analysis.
- To provide suitable treatment and preventive recommendations to help farmers take timely action.

Proposed Solution

The proposed system provides a mobile-based solution through which farmers can capture a clear image of an affected crop or leaf. The captured image is first passed through an image preprocessing stage where noise is reduced, brightness is adjusted, the image is resized, and important visual features are enhanced.

The processed image is then analyzed by a trained Deep Learning model. The model identifies important characteristics such as colour changes, spots, textures, patterns, and other visual symptoms associated with crop diseases. Environmental information such as temperature and humidity can also be integrated through an API to provide additional context for prediction.

After analysis, the system displays the detected disease, confidence score, possible causes, and recommended treatment or preventive measures. This allows farmers to identify potential problems at an early stage and take appropriate action, thereby reducing crop losses and supporting better agricultural productivity.

Technologies Used

Category	Technology
Mobile Application	Android Studio
Programming Language	Kotlin / Java
User Interface	XML, Material Design
Machine Learning	Python
Deep Learning	TensorFlow / Keras
Model	CNN / MobileNetV2 / EfficientNet
Computer Vision	OpenCV
AI Deployment	TensorFlow Lite (TFLite)
Backend	Python Flask
Database	MySQL / SQLite
API Integration	Weather API
Data Processing	NumPy, Pandas
Data Visualization	Matplotlib
Model Training	Google Colab / Jupyter Notebook
Dataset	PlantVillage / Custom Crop Disease Dataset
Version Control	Git / GitHub

Social Impact (To whom your Project Benefits)

The primary beneficiaries of this project are farmers, especially those who may not have immediate access to agricultural experts. The system can help farmers detect crop diseases more quickly and make informed decisions regarding treatment and prevention.

Early disease detection can help reduce crop losses, minimize unnecessary use of pesticides, improve crop quality, and increase agricultural productivity. Agricultural advisors and organizations can also use such technology as a supporting tool for crop-health monitoring. Overall, the project contributes to smart and technology-driven agriculture by making AI-based crop health analysis more accessible.

Conclusion

The AI-Based Crop Disease Detection and Recommendation System provides an intelligent approach to crop health monitoring by combining Computer Vision, Deep Learning, mobile technology, and environmental data. By allowing farmers to capture a crop image and receive an automated disease prediction with recommendations, the system can make disease identification faster and more accessible.

The proposed solution aims to support early disease detection, timely treatment, reduced crop losses, and improved productivity. It demonstrates how Artificial Intelligence can be applied in AgriTech to address real-world agricultural challenges and promote smarter and more sustainable farming practices.